Enterprise MCQs (Advanced Scenario-Based – No Answers)

🔹 Case 1: Financial Data Processing System

A multinational bank processes millions of transactions daily. During testing, a developer writes:

```
amount = "10000"
tax = 0.18
total = amount * tax
```

The system produces incorrect results in production.

Q1. What is the most likely issue in this implementation?

A. Incorrect multiplication logic
B. Type mismatch leading to unintended behavior
C. Syntax error in Python
D. Memory allocation issue

🔹 Case 2: Telecom Data Deduplication

A telecom company receives customer phone numbers from multiple sources. Duplicate entries must be removed efficiently while ensuring fast lookup performance.

Q2. Which data structure is most suitable for this requirement?

A. List
B. Set
C. Tuple
D. String

🔹 Case 3: Retail ETL Pipeline

An ETL job processes millions of sales records. Some records are corrupted and must be skipped without interrupting the pipeline.

Q3. Which control statement should be used?

A. break
B. continue
C. pass
D. return

🔹 Case 4: Fraud Detection Engine

A fraud detection system must stop processing immediately when a suspicious transaction exceeds a threshold.

Q4. Which control statement is most appropriate?

A. continue
B. break
C. pass
D. return

🔹 Case 5: Microservices Architecture

A logistics platform uses reusable Python functions for pricing and route optimization across services.

Q5. Why are functions critical in this architecture?

A. Reduce memory usage
B. Improve modularity and reusability
C. Replace APIs
D. Eliminate dependencies

🔹 Case 6: Real-Time Data Transformation

A streaming system requires lightweight inline transformation of incoming data.

Q6. Which approach is most suitable?

A. lambda x: return x+1
B. lambda x: x+1
C. def lambda(x): x+1
D. lambda(x): x+1

🔹 Case 7: Cybersecurity Log Analysis

A system scans logs to extract all email IDs and IP addresses for threat detection.

Q7. Which regex method is most suitable?

A. match()
B. findall()
C. search()
D. replace()

🔹 Case 8: Distributed Logging System

An enterprise application writes logs continuously across multiple servers. Logs must not overwrite previous entries.

Q8. Which file mode should be used?

A. 'w'
B. 'a'
C. 'r'
D. 'x'

🔹 Case 9: Payment Gateway Stability

A payment API must handle invalid inputs without crashing.

Q9. What is the best approach?

A. Ignore invalid inputs
B. Use structured error handling
C. Print errors only
D. Restart service

🔹 Case 10: Banking Error Handling

A banking system defines a custom exception `TransactionLimitExceeded`.

Q10. Why is this approach used?

A. Improve performance
B. Handle domain-specific errors clearly
C. Replace syntax errors
D. Reduce memory usage

🔹 Case 11: Healthcare Management System

A hospital system models patients, doctors, and appointments using Python classes.

Q11. Why is OOP suitable for this scenario?

A. Reduces execution time
B. Models real-world entities effectively
C. Improves UI
D. Eliminates need for functions

🔹 Case 12: Shared Configuration

A system needs a tax rate shared across all instances of a class.

Q12. Which variable type should be used?

A. Instance variable
B. Class variable
C. Local variable
D. Temporary variable

🔹 Case 13: Payment Gateway Integration

A payment system supports multiple payment types using the same interface but different implementations.

Q13. Which concept is applied?

A. Encapsulation
B. Polymorphism
C. Inheritance
D. Aggregation

🔹 Case 14: Dynamic Method Behavior

Different subclasses modify the same method based on business logic.

Q14. What is this concept called?

A. Overloading
B. Overriding
C. Abstraction
D. Composition

🔹 Case 15: Corporate Communication

A consultant sends an update email to stakeholders including senior management.

Q15. What is the best communication approach?

A. Informal tone
B. Structured and professional communication

C. Long unstructured paragraphs
D. No subject line

🔹 Case 16: Client Onboarding

During the first meeting with a new client, expectations must be aligned.

Q16. What should be done first?

A. Technical discussion
B. Ice breaking and objective setting
C. Code demonstration
D. Reporting

🔹 Case 17: Hierarchical Data Processing

A system processes organizational hierarchy using recursion.

Q17. What is essential for recursion to function correctly?

A. Loop statement
B. Base condition
C. Global variable
D. Exception handling

🔹 Case 18: Data Conversion Pipeline

User input from a web application is received as string but needs numeric processing.

Q18. What is the correct approach?

A. str()
B. float()
C. eval()
D. input()

🔹 Case 19: Configuration Storage

A system stores configuration values that must remain immutable and ordered.

Q19. Which data structure is most suitable?

A. List
B. Tuple
C. Set
D. Dictionary

🔹 Case 20: Production Deployment

A Python script is deployed in production and must avoid unintended execution when imported as a module.

Q20. What is the correct practice?

A. Use global variables
B. Use main guard (`if __name__ == "__main__":`)
C. Use print statementsA
D. Avoid functions